

PDF ANONIMIZADO

ID publico: case_0267

Paginas del documento: 10

Copia anonimizada para verificacion metodologica.

Se removieron portada editorial, titulo, autores, afiliaciones, correos, DOI, URLs y metadatos del PDF original.

Las paginas restantes conservan el contenido util para revisar metodos, resultados, tablas y conclusiones.

Nota: la anonimización automatica prioriza reducir la identificacion directa del articulo. Los casos de una sola pagina o diseno no convencional deben revisarse manualmente antes de una publicacion abierta.

ness of educational initiatives, an integrated multidisciplinary approach involving health actors, and parents/guardians is essential [2,3]. Schools play a pivotal role in health promotion, offering opportunities to collectively reach children, educators, and families. Public policies increasingly emphasize the integration of the education and health sectors to support comprehensive well-being. In this context, Oral Health Teams' involvement within the Family Health Strategy and the School Health Program has gained prominence [4]. Since the mid-20th century, oral health promotion has been widely implemented and recognized within school communities [5].

Preventive measures are conducted daily or weekly to reduce the risk of tooth decay and reinforce oral hygiene habits among schoolchildren, with active participation from educators [6], such as indirectly supervised brushing with fluoridated toothpaste. For preschoolers, using fluoridated toothpaste containing a conventional fluoride concentration (1000 to 1500 µg/g) in reduced amounts is recommended as of the eruption of their first tooth [7,8]. During early childhood, specific oral care is crucial due to the ongoing mineralization of permanent dental crowns. Appropriate fluoridated toothpaste prevents dental caries while minimizing the risk of dental fluorosis [9]. Adult supervision is essential to ensure proper application and rinsing since young children tend to swallow a significant amount of toothpaste during brushing [10-13]. Additionally, caregivers should apply toothpaste to the child's toothbrush and encourage expectoration after brushing [14].

Although educators are generally committed to their students' oral health and feel responsible for it, they report a need for training to address these issues safely, underscoring the importance of educational initiatives targeting teachers [15-21]. In this context, promoting actions focused on children's safe use of toothpaste is essential for optimizing caries prevention and minimizing the risk of dental fluorosis. Educational tools, such as validated comic books, have shown promise in improving health literacy [22] by facilitating the comprehension of complex concepts [23]. While often regarded as playful and accessible, comic books can also serve as practical instructional tools [23], demonstrating superior efficacy in conveying health knowledge compared to traditional methods [24,25].

The validated educational comic book titled *Dentitos in Fluoride and Oral Health* [26] has proven effective in guiding adults on the safe use of fluoridated toothpaste for children [27]. As a passive self-instructional resource, it has enhanced knowledge on indirectly supervised tooth brushing and contributed to the professional development of educators as agents of oral health promotion [28]. Exploring alternative strategies for utilizing this comic book could further strengthen educators' knowledge, improving its application in educational settings. However, the lack of evidence comparing the effectiveness of different methods of using the comic book highlights the need for studies such as the present one. Hence, this study aimed to evaluate various methods of using a validated comic book as an educational strategy to promote the safe use of fluoridated toothpaste among children.

■ Materials and Methods

Ethical Clearance and Study Design

This quasi-experimental study complied with Resolution #466 of 2012 of the National Health Council, Ministry of Health, and was approved by the local research ethics committee (Opinion no. 5.884.353). Informed consent was obtained from all educators before participation, ensuring they understood the study's purpose, the

was conducted in Macapá, the capital of Amapá, Brazil, in 2022. Macapá has a population of 933, with 58.5% residing in areas without access to treated and fluoridated water [30].

Randomization and Intervention

A cluster randomization design was employed, with schools as the unit of randomization. Schools were randomly assigned to groups using a random number generator to ensure unbiased allocation:

material independently without researcher supervision.

participated in a group discussion facilitated by a dentist, which covered the content of the comic book and its application in oral health practices and addressed participants' questions.

Participants

The study included educators recruited from public schools in Macapá. Eligible participants were required to be actively engaged in teaching and have a basic understanding of oral health practices. A significance level of 5% and a test power of 80% were considered for sample size determination. The selected sample size provided 80% power to detect effect sizes of $d=1.02$ for intergroup comparisons and $d_z=0.81$ for intragroup comparisons. Educators who completed only one questionnaire or failed to provide identifying information were excluded to ensure paired comparisons.

Outcome Measures

The primary outcome was the educators' knowledge and attitudes toward oral health practices, which were assessed using a validated questionnaire of 2 open-ended and 27 closed-ended questions [28]. The questionnaire collected demographic data, details on routine school oral health activities, prior knowledge of fluoride use in children, and awareness of dental fluorosis. It also included Likert-scale items evaluating the perceived importance and risk of fluoridated toothpaste use in children, with five situations rated on a scale from 0 (no importance) to 5 (total importance) and five situations rated from 0 (lowest risk) to 5 (highest risk). Additionally, to assess their practical understanding, educators were asked to illustrate the amount of toothpaste they would use during supervised brushing, thus visually representing their knowledge of appropriate toothpaste application [28].

The questionnaires were administered before the intervention (T0) and one week after the final meeting with each group (T1). Changes in responses between T0 and T1 were analyzed to evaluate the impact of the intervention.

Data Collection and Analysis

Data from the questionnaires were collected and stored securely for analysis, which was conducted by researchers blinded to group allocation. Categorical variables were summarized using absolute and relative frequencies, while continuous variables were described using means, standard deviations, medians, and range values. The size of the toothpaste illustration drawn by the educators was measured in millimeters using a

agroup comparisons across different time points. A significance level of 5% was adopted for es, which were performed using the R programming language.

■ Results

Of the 45 educators invited to participate in the study, seven completed only one questionnaire, and five did not provide identifying information, leading to their exclusion. Although these exclusions reduced the final sample size, the remaining participants ensured sufficient statistical power for analysis. The final sample consisted of 33 educators, with 19 in G1 and 14 in G2. Randomization was conducted at the school level, with entire schools randomly assigned to either group. The interventions were then delivered to groups of educators within these schools, maintaining the cluster design throughout the study.

Table 1 presents descriptive data on the participants' profiles and the oral healthcare practices implemented in schools. Most educators in both groups had completed college, worked with children up to 4 years old, and participated in supervised toothbrushing activities at school. All schools had appropriate facilities for these activities.

Table 1. Descriptive data on participants' profiles and oral healthcare provided in schools.

Variables	Group	
	G1 N (%)	G2 N (%)
Age (years)		
Mean age (standard deviation)	38.6 (11.0)	43.8 (11.5)
Minimum and maximum age	20.8 to 63.0	18.3 to 59.5
Schooling [n (%)]		
Incomplete College	0 (0.0)	1 (7.1)
Completed College	19 (100.0)	13 (92.9)
Age of the majority of the children you work with		
Less than 3 years	2 (10.5)	0 (0.0)
3 to 4 years	8 (42.1)	11 (78.6)
5 years	9 (47.4)	3 (21.4)
Do children brush their teeth at school?		
They never do	1 (5.3)	0 (0.0)
Sometimes they do	6 (31.6)	6 (42.9)
Yes, they do it every day	12 (63.2)	8 (57.1)
The school has a designated and suitable area for children to brush their teeth.		
Yes	19 (100.0)	14 (100.0)

Changes in Knowledge and Perceptions

Table 2 compares participants' responses to two key questions at different times. In G1, participants shifted their responses to "Can small children use toothpaste with fluoride?" from "no" to "yes," and from "yes" to "yes, as long as it is children's toothpaste." In G2, responses evolved from "yes, as long as it is children's toothpaste" to simply "yes," indicating that their understanding of the subject was such that children could use the same toothpaste as adults. For the question "Do you know the ideal amount of toothpaste to apply to a small child's toothbrush?" notable response shifts occurred. In G1, 31.6% of educators changed their responses from "I'm not sure" to "Yes" after reading the comic book. In G2, 64.3% of participants changed their responses from "no" (14.3%) or "I'm not sure" (50.0%) to "yes" after the intervention.

Questions	Group	Before Reading	No	After Reading	
			N (%)	Yes	Yes, as long as it is children's toothpaste
Can young children use fluoride toothpaste?	1	No	0 (0.0)	1 (5.3)	0 (0.0)
		Yes	0 (0.0)	7 (36.8)	1 (5.3)
		Yes, as long as it is children's toothpaste	0 (0.0)	2 (10.5)	8 (42.1)
	2	No	0 (0.0)	0 (0.0)	0 (0.0)
		Yes	0 (0.0)	3 (21.4)	0 (0.0)
		Yes, as long as it is children's toothpaste	0 (0.0)	8 (57.1)	3 (21.4)
Do you know the ideal portion of toothpaste for a small child's toothbrush?	1	No	0 (0.0)	I am not sure	Yes
		I am not sure	0 (0.0)	0 (0.0)	6 (31.6)
		Yes	0 (0.0)	1 (5.3)	12 (63.2)
	2	No	0 (0.0)	0 (0.0)	2 (14.3)
		I am not sure	0 (0.0)	0 (0.0)	7 (50.0)
		Yes	0 (0.0)	0 (0.0)	5 (35.7)

Changes in the Amount of Toothpaste Applied

Table 3 presents data on participants' drawings representing the amount of toothpaste they would use or currently use on children's toothbrushes at their respective preschools, categorized by group and time point. Both groups significantly reduced the amount of toothpaste applied ($p < 0.05$). In the first questionnaire, participants illustrated the amount of toothpaste they would use or currently use, and the examiner measured these drawings in millimeters to calculate the median values. The same procedure was repeated in the second questionnaire, revealing a marked reduction in the size of the amount drawn, indicating greater adherence to recommended toothpaste quantities.

Table 3. The drawings' measurements (in millimeters) represent the amount of toothpaste that educators would use or currently use on children's toothbrushes at school, categorized by group and time point.

Group	Before Reading		After Reading		p-value*
	Mean (SD)	Median (Min. and Max.)	Mean (SD)	Median (Min. and Max.)	
1	7.0 (2.4)	7.0 (3.0; 11.0)	3.4 (1.7)	3.0 (2.0; 8.0)	0.0001
2	6.9 (2.2)	6.0 (3.0; 12.0)	2.8 (0.8)	3.0 (2.0; 4.0)	0.0010
p-value*		0.8555		0.5004	

*Comparison between each group's time points (before and after reading the comic); *Comparison between groups at each time point; SD: Standard Deviation; Min.: Minimum; Max.: Maximum.

ge perception scores across groups and time points. In G2, there was a significant increase in scores related to knowledge of the appropriate , the importance of care since children are not responsible for applying toothpaste, the necessity of adult supervision, and encouragement of expectoration after brushing ($p<0.05$). In both groups, perception scores increased significantly for the risk of "the child having toothpaste on their toothbrush while waiting for their turn to brush at school" ($p<0.05$).

Table 4. Knowledge score analyses for topics covered in the questionnaire, categorized by group and time point.

Topic	Group	Before Reading		After Reading		p-value*
		Mean (SD)	Median (Min.; Max.)	Mean (SD)	Median (Min.; Max.)	
An adult applying toothpaste to children's toothbrushes	1	4.5 (0.8)	5.0 (3.0; 5.0)	4.8 (0.5)	5.0 (3.0; 5.0)	0.1508
	2	4.5 (0.7)	5.0 (3.0; 5.0)	4.9 (0.3)	5.0 (4.0; 5.0)	0.0759
p-value*		0.8179		0.8555		
Paying attention to the amount of toothpaste based on the child's age	1	4.4 (0.9)	5.0 (3.0; 5.0)	4.7 (0.7)	5.0 (3.0; 5.0)	0.1056
	2	4.5 (0.5)	4.0 (4.0; 5.0)	4.9 (0.3)	5.0 (4.0; 5.0)	0.0277
p-value*		0.7012		0.6489		
Monitoring whether the child spits out the toothpaste while brushing	1	4.7 (0.7)	5.0 (2.0; 5.0)	4.8 (0.5)	5.0 (3.0; 5.0)	0.5930
	2	4.3 (1.0)	5.0 (2.0; 5.0)	4.6 (0.6)	5.0 (3.0; 5.0)	0.4413
p-value*		0.2658		0.4021		
Encouraging the child to spit out toothpaste during and after brushing	1	4.8 (0.5)	5.0 (3.0; 5.0)	4.8 (0.5)	5.0 (3.0; 5.0)	0.7893
	2	4.5 (0.8)	5.0 (3.0; 5.0)	4.9 (0.3)	5.0 (4.0; 5.0)	0.0935
p-value*		0.2742		0.8555		
The child has toothpaste on their toothbrush while waiting to brush their teeth at school	1	3.0 (1.6)	3.0 (1.0; 5.0)	3.7 (1.5)	4.0 (1.0; 5.0)	0.0180
	2	3.3 (1.2)	3.0 (2.0; 5.0)	4.6 (0.8)	5.0 (3.0; 5.0)	0.0254
p-value*		0.7220		0.1451		
The child applies toothpaste to their toothbrush	1	3.8 (1.5)	4.0 (1.0; 5.0)	4.3 (1.3)	5.0 (1.0; 5.0)	0.1141
	2	3.8 (1.3)	4.0 (2.0; 5.0)	4.9 (0.3)	5.0 (4.0; 5.0)	0.0251
p-value*		0.9681		0.2155		
An adult supervising children's brushing	1	2.9 (1.9)	3.0 (1.0; 5.0)	2.4 (1.9)	1.0 (1.0; 5.0)	0.2626
	2	3.3 (1.6)	3.5 (1.0; 5.0)	4.7 (0.6)	5.0 (3.0; 5.0)	0.0284
p-value*		0.5952		0.0016		
The child was swallowing toothpaste while brushing	1	4.1 (1.5)	5.0 (1.0; 5.0)	4.8 (0.7)	5.0 (2.0; 5.0)	0.1013
	2	3.8 (1.6)	5.0 (1.0; 5.0)	4.9 (0.3)	5.0 (4.0; 5.0)	0.0425
p-value*		0.5649		0.8780		
An adult supervises the child to encourage spitting out toothpaste	1	2.7 (1.8)	2.0 (1.0; 5.0)	3.1 (2.1)	5.0 (1.0; 5.0)	0.7353
	2	2.9 (1.4)	3.0 (1.0; 5.0)	4.9 (0.4)	5.0 (4.0; 5.0)	0.0152
p-value*		0.6264		0.0535		

*Comparison between each group's time points (before and after reading the comic); *Comparison between groups at each time point; SD: Standard Deviation; Min.: Minimum; Max.: Maximum.

ting the effectiveness of comic books as educational tools in oral health [23,28]. Additionally, group discussion mediated by a dentist further enhanced the observed benefits, particularly by increasing accurate knowledge about the appropriate use of toothpaste, the ideal amount for children, and the risks of dental fluorosis. These findings support that combining passive educational tools with interactive professional guidance can optimize educational outcomes.

In school settings, educators are responsible for conducting indirectly supervised toothbrushing [6]. Therefore, strategies that promote their autonomy in performing this health-related task are essential for maximizing benefits while minimizing potential risks [31]. The participating schools had designated areas for children to brush their teeth, facilitating daily oral hygiene routines. Since more than half of Macapá's population lacks access to fluoridated water, school-based toothbrushing programs become even more critical. In this context, schools are ideal environments for implementing health programs that can significantly influence children's oral hygiene practices and foster long-lasting healthy habits. Educational and preventive programs in schools have been shown to reduce dental caries, particularly in regions with limited access to fluoride [6], such as Macapá, where such interventions help bridge gaps in oral health care.

Most educators in the study had completed college, suggesting a high level of health literacy that enables them to make informed decisions regarding the oral health of the children in their care [32]. This may explain why, even before reading the comic book, most educators knew that young children should use fluoridated toothpaste and understood the appropriate amount for brushing. However, some participants in G2 expressed uncertainty about these topics, underscoring the need to reinforce these concepts further.

This study focused on children under five, a critical period for establishing positive health behaviors. Early oral health interventions are essential for fostering proper attitudes toward oral hygiene [33]. However, early exposure to fluoride-containing toothpaste increases the risk of dental fluorosis, particularly in young children, owing to the higher likelihood of accidental ingestion during enamel maturation [34]. Therefore, fluoridated toothpaste must be carefully monitored in preschool settings to balance caries prevention and the risk of fluorosis [35].

In general, the results from G1 demonstrated that reading the comic book alone improved knowledge. This finding aligns with previous studies assessing the passive use of educational comic books in the home [27] and school [28] settings. The ability to use the comic book independently, without requiring a health professional, enhances its potential for widespread implementation in schools. This underscores the critical role of educators in delivering such interventions [37].

However, knowledge improved significantly when a health professional facilitated a group discussion following independent reading. This was particularly evident in G2, where participants better understood the appropriate use of fluoridated toothpaste, the correct amount to apply, and the risks associated with dental fluorosis. These findings highlight the value of combining educational materials with professional guidance to enhance the effectiveness of health interventions.

Placing small amounts of fluoridated toothpaste at the ideal concentration (1000–1500 ppm F), supervising brushing, and encouraging expectoration are highly recommended practices [38], all emphasized in the comic book. This is particularly relevant in regions such as Amapá, where dental caries remain a significant

manent teeth, particularly in children under four years of age [31]. In both groups, the percentage of participants intended to apply fluoride to children's toothbrushes decreased significantly. Moreover, awareness notably increased regarding the risk of the child inadvertently ingesting fluoride from the toothpaste on their brush while waiting to brush ($p < 0.05$). These findings highlight the importance of reinforcing proper toothpaste application in supervised brushing routines at preschools. Since fluoride ingestion during early childhood can lead to dental fluorosis and aesthetic impairments in permanent teeth, ensuring adherence to recommended practices is essential, particularly for children under four years of age [26].

One limitation of this study is that the acquired knowledge may not always translate into expected behavioral changes, given the demanding school routine and imbalance between the number of educators and students. Additionally, as a quasi-experimental study, the lack of complete randomization at the individual educator level may limit the generalizability of the findings. Potential confounding factors were not fully controlled, which should be considered when interpreting the results. Nonetheless, this study demonstrates that educational materials can effectively enhance the knowledge of non-expert individuals regarding the safe use of fluoridated toothpaste in non-clinical settings.

Future studies should investigate the effectiveness of supervised toothbrushing in school environments, focusing on key aspects, such as comprehensive professional training, access to appropriate materials, correct toothpaste application, and adequate supervision during brushing activities. Additionally, research should examine the long-term retention of knowledge gained through educational interventions and how increased awareness translates into consistent and effective daily practices within and beyond the school setting.

Schools play a pivotal role in promoting children's oral health since children spend a significant portion of their time in this environment. Thus, fostering health literacy at an early stage is essential [39]. The findings of this study underscore the importance of collaboration between the education and healthcare sectors to enhance oral health outcomes and ensure the safe implementation of preventive measures.

■ Conclusion

Reading the comic book alone effectively conveyed knowledge about using fluoridated toothpaste safely. However, when supplemented with a group discussion, the intervention significantly enhanced participants' understanding of safe fluoride use in children. These findings highlight the value of combining independent reading with interactive discussions to maximize the impact of educational strategies in promoting safe oral health practices.

■ Authors' Contributions

■ Financial Support

None.

ity

■ References

- [1] Mishra A, Pandey RK, Chopra H, Arora V. Oral health awareness in school-going children and its significance to
antes e após a política nacional de saúde bucal. *Rev Bras Pesqui Saúde* 2019; 21(3):30-38. [In Portuguese].
- [6] Carvalho KN, Zanin L, Flório FM. Students and nurses perception regarding the educational practices of the health at
Brasil. Brasília: Ministério da Saúde; 2009. [In Portuguese].
- [8] Cury JA, Tenuta LM. Evidence-based recommendation on toothpaste use. *Braz Oral Res* 2014; 28 Spec No:1-7.
- editores. *Odontopediatria para pediatria*. 1st ed. São Paulo: Atheneu; 2013. p. 419-30. [In Portuguese].
- [11] Lima YB, Cury JA. Ingestão de flúor por crianças pela água e dentifríco. *Rev Saude Publica* 2001; 35(6):576-581.
- year-old US children. *Community Dent Health* 2014; 31(3):136-140.
- [16] Santos PA, Rodrigues JDA, Garcia PPNS. Avaliação do conhecimento dos professores do ensino fundamental de escolas
particulares sobre saúde bucal. *Rev Odontol UNESP* 2002; 31(2):205-214. [In Portuguese].
- [17] Franchin V, Basting RT, Mussi ADA, Flório FM. A importância do professor como agente multiplicador de saúde
bucal. *Rev Abeno* 2006; 6(2):102-108. [In Portuguese].
- [18] Granville-Garcia AF, Silva JM, Guinho SF, Menezes V. Conhecimento de professores do ensino fundamental sobre
saúde bucal. *RGO* 2007; 55(1):29-34. [In Portuguese].
- [19] Vasel J, Bottan ER, Campos L. Educação em saúde bucal: Análise do conhecimento dos professores do ensino
fundamental de um município da região do Vale do Itapocu (SC). *RSBO* 2008; 5(2):12-18. [In Portuguese].
- elementary school teachers from a small city in the savanna region of the state of Paraíba, Brazil. *Arq Odontol* 2014;
50(4):193-202. [In Portuguese].
- [22] Pereira G, Zanin L, Flório FM. Prechool teachers of Santos - Brazil: What do they know about oral health? *Contexto*

Leopoldo Mandic; 2021. [In Portuguese].

[28] Flório FM, Rached EA, Victorelli G, Silva ASF, Arsati YBOL. Development and validation of an educational comic

Nacional de Saúde Bucal: resultados principais. Brasília: Ministério da Saúde; 2012. [In Portuguese].

[38] Flório FM, Oliveira A de, Fonseca Silva A de S. Workshops from the "Playing and Learning about Health" project:

IL: American Academy of Pediatric Dentistry; 2023. 3528p.

[40]